제2장 전력 계산

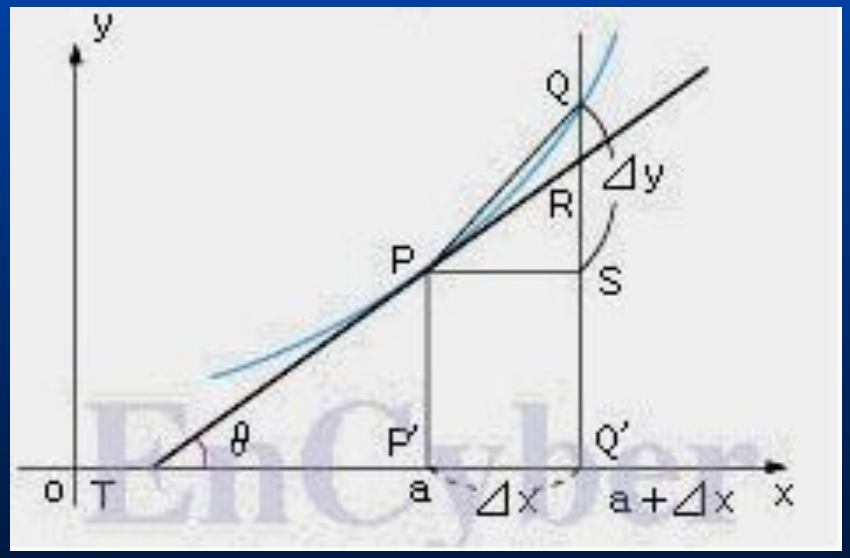
Power Computation

2.1 서론

- 전기의 기본 개념 정리
- 전력계산
  정현파(Sin Wave)
  비정현파
- OrCAD를 이용하여 전력계산

예비학습: 미분(differentiation)

- (x)가 미분 가능인 경우에 y=f(x)라 놓고 x와 y의 증분을 각각 Δx,Δy로 놓고 변화율을 구함

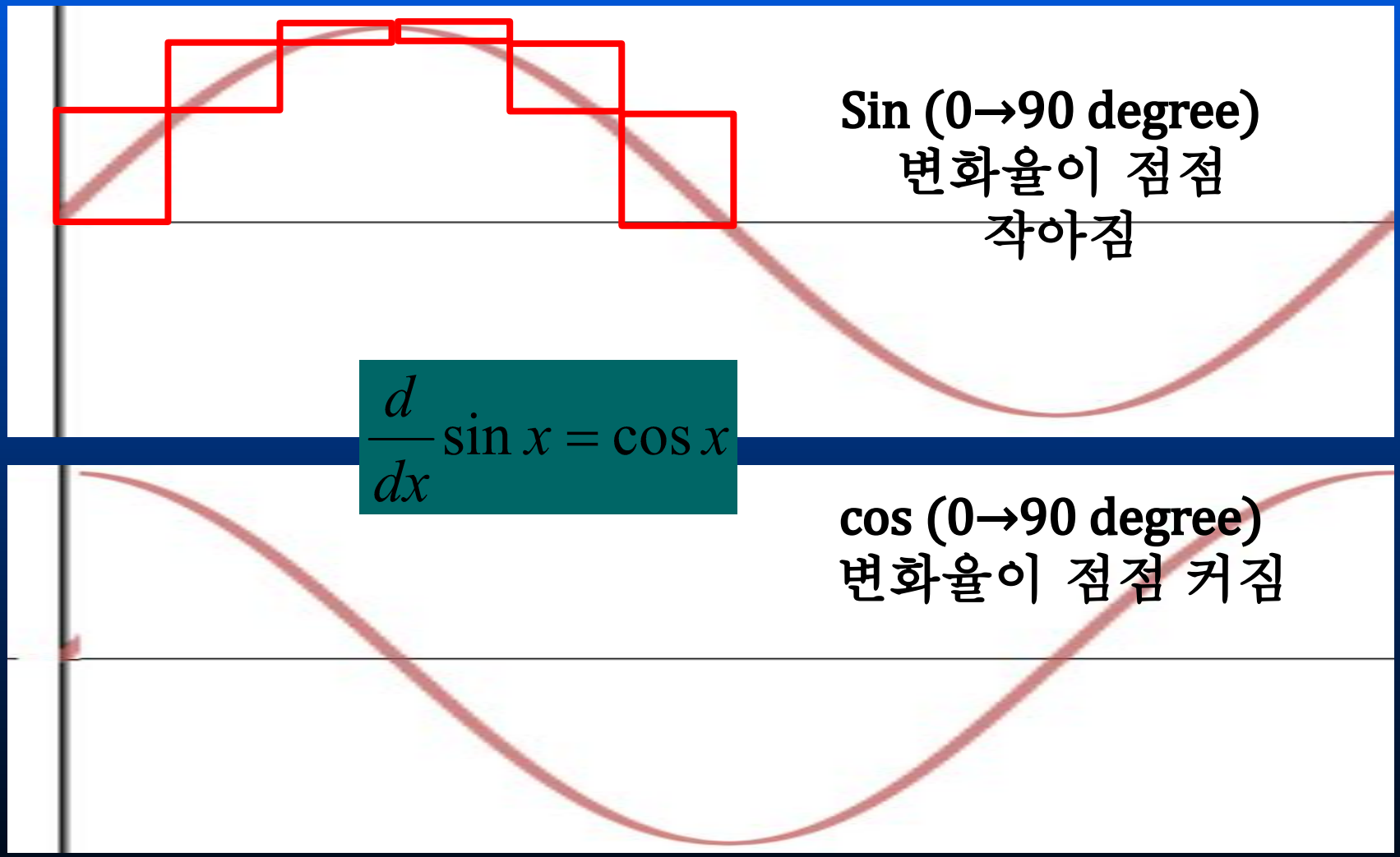


$$f'(x) = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{[f(x+\Delta x) - f(x)]}{\Delta x}$$

$$= \frac{dy}{dx}$$

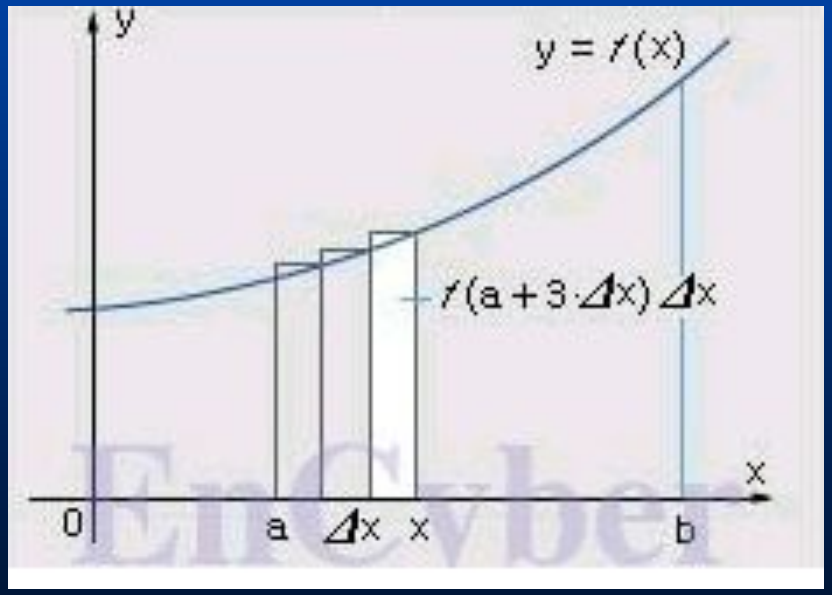
미분(differentiation)=변화율



$$\frac{d}{dx}\sin x = \cos x$$

예비학습: 적분(integral)

- 곡선 f(x)가 있을 때, 이 곡선과x축 및 두 직선 $x_1 = a$, $x_2 = b(b > a)$로 둘러싸인 평면의 넓이를 구하면



$$\lim_{n \to 0}\left[\sum_{k=1}^{n} f(a + k\Delta x)\right]\Delta x$$

$$F'(x) = f(x)$$

$$\int_a^b f(x)dx = F(X)$$

적분(integral)=넓이

f 가로=3 세로=2 d• =

3×2=6

f 반지름=3 d ○ = 3 × 3 × π = 9π

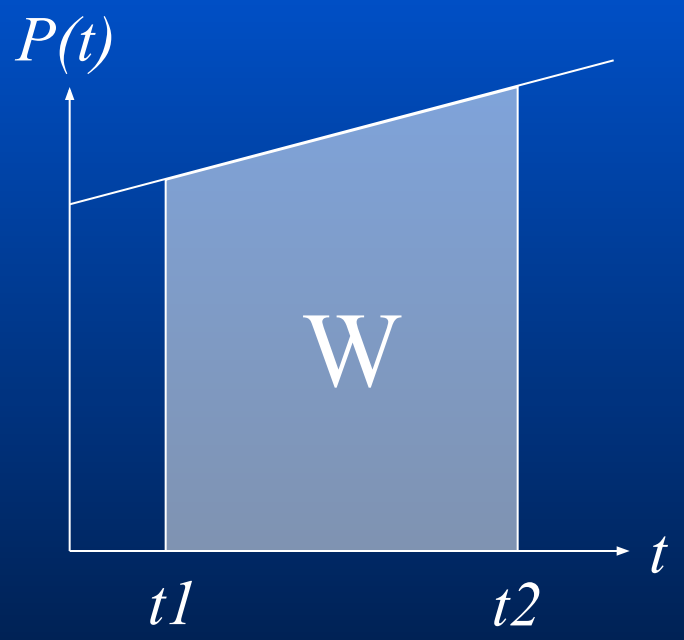
f d■ =공식

$$\int 1\,dt = t$$

$$\int x\,dx = \frac{1}{2}x^2$$

$$\int y^{100}\,dy = \frac{1}{101}y^{101}$$

2.2 전력과 에너지



- 순시전력 (Instantaneous power)

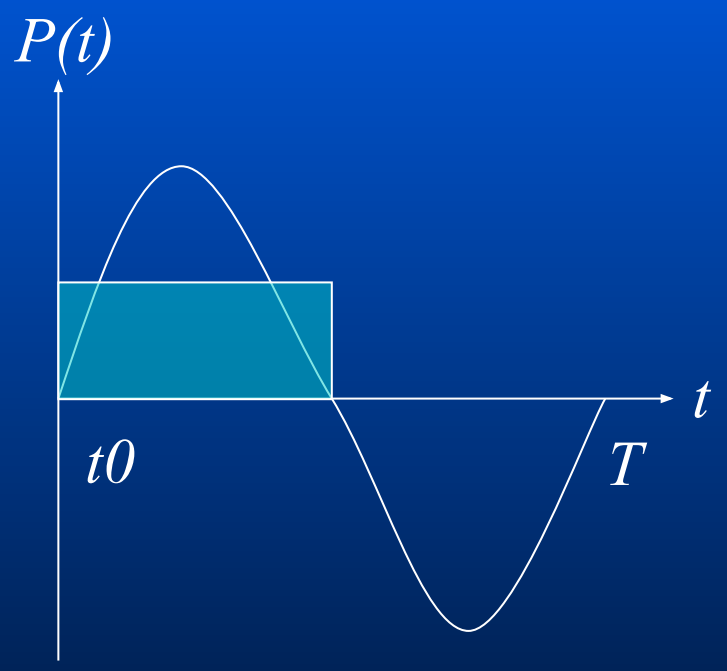
$$p(t) = i(t)v(t)\ [W]$$

- 에너지(Energy)

$$w(t) = \int_{t1}^{t2} p(t)\,dt\ [J]$$

→ *joule[J]= watt [W]× time[s] [KWh]*

2.2 전력과 에너지: 평균값(Mean Value)



*정현파는 반주기로 계산
한주기 계산시 0가 됨

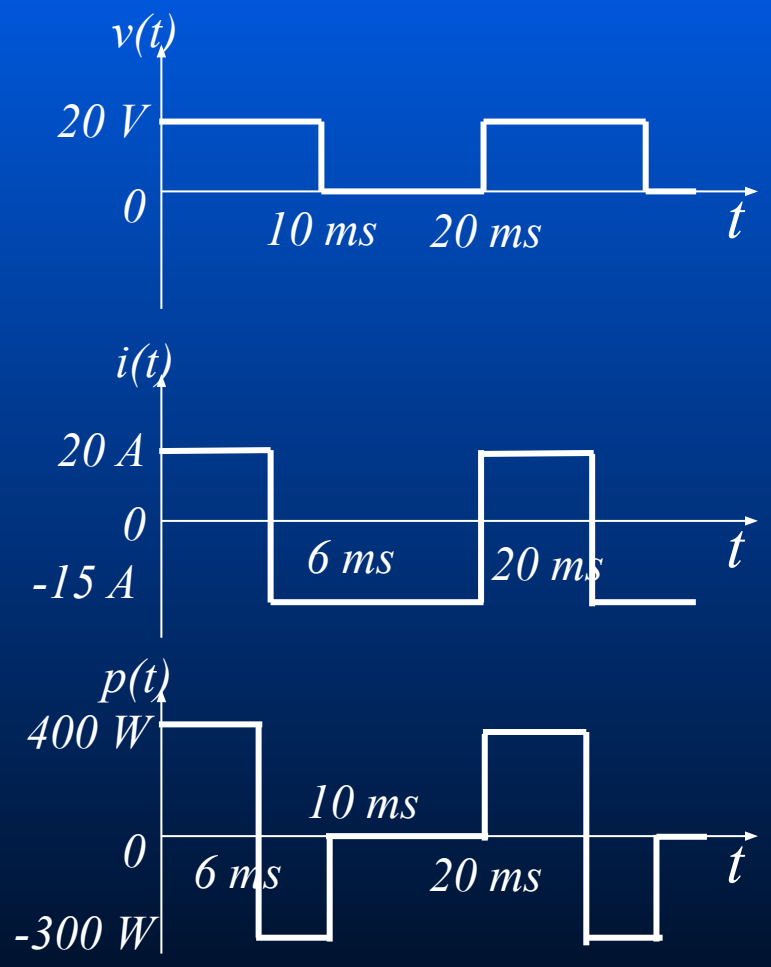
- 전력의 평균값

$$P = \frac{1}{T}\int_{t0}^{t0+T} p(t)\,dt$$

$$= \frac{1}{T}\int_{t0}^{t0+T} v(t)i(t)\,dt$$

$T =$ 주기(period)

예제 2-1 전력과 에너지



- 에너지

$$W = \int_{t0}^{t0+T} p(t)\,dt$$

$$= \int_{0}^{0.006} 400\,dt + \int_{0.006}^{0.01} -300\,dt + \int_{0.01}^{0.02} 0\,dt$$

$$= [400t]_{0}^{0.006} + [-300t]_{0.006}^{0.01}$$

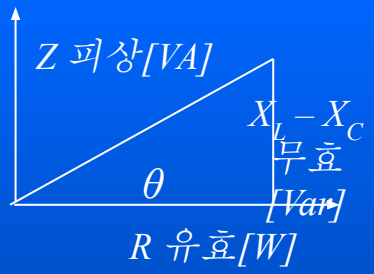
$$= (2.4-0)+(-3+1.8) = 1.2[J]$$

- 평균전력

$$P = \frac{1}{T}\int_{t0}^{t0+T} p(t)\,dt$$

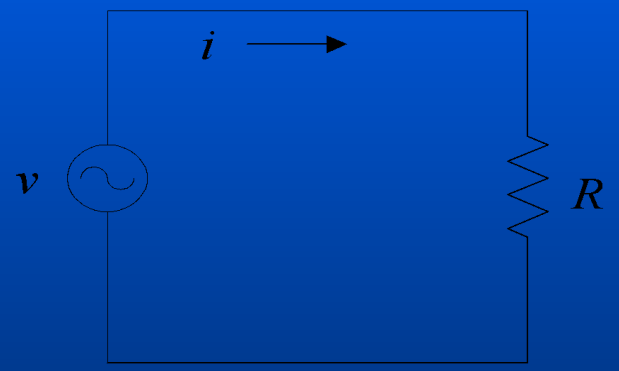
$$= \frac{1}{0.02}[1.2] = 60[W]$$

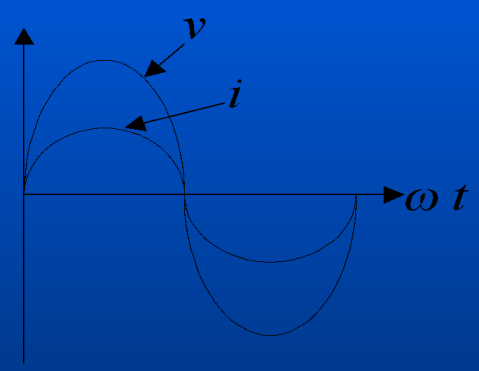
중요: 수동소자



직류 (주파수 f =0)			교류 (주파수 f >0)		
	기본공식	에너지저장		리액턴스	임피던스/역율
R 저항 [Ω]	$I = \frac{V}{R}$	0	R [Ω]	R	$Z = \sqrt{R^2 + (X_L - X_C)^2}$ $cos\theta = \frac{R}{Z} = \frac{유효}{피상}$
L 인덕턴스 [H]	$V = L\frac{di}{dt}$	$W = \frac{1}{2}Li^2$	X_L [Ω]	$X_L = 2\pi f L$ 유도성	
C 커패시턴스 [F]	$i = C\frac{dv}{dt}$	$W = \frac{1}{2}Cv^2$	X_C [Ω]	$X_C = \frac{1}{2\pi f C}$ 용량성	

예비학습: 저항(R) 회로



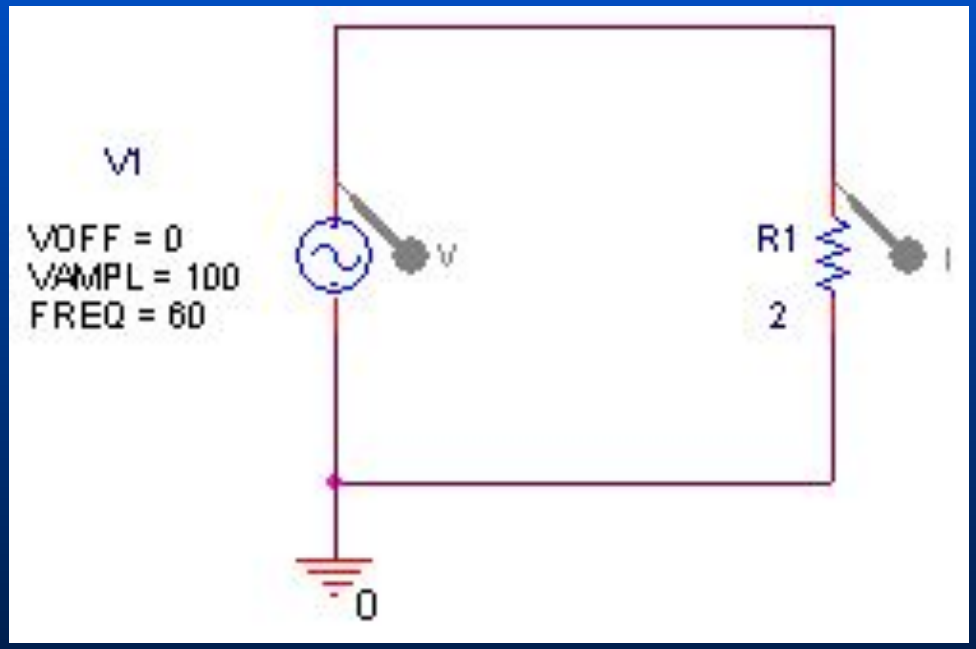


- 전류

$$i = \frac{v}{R}$$

- 전압과 전류의 위상: 동상(same phase)

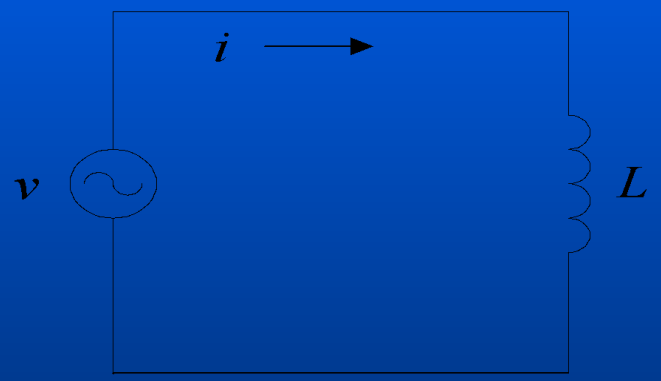
R 회로: OrCAD Simulation

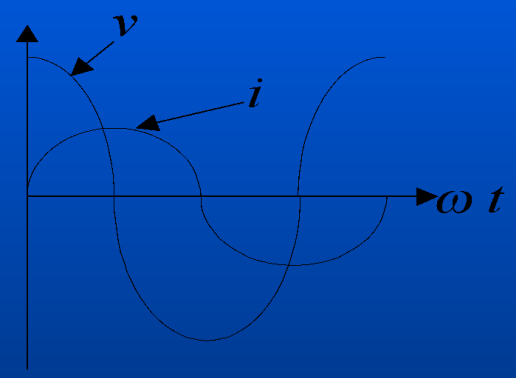


- *Parameter*
  - ✔ *R=2 Ω*
  - ✔ *V1=100 V*
  - ✔ *f=60 Hz*
  - ✔ *Transient Step:*

    0 0.1 ms 50 ms

예비학습: 인덕터(L) 회로





$$\frac{d}{dx}\sin x = \cos x$$

$$\frac{d}{dx}\cos x = -\sin x$$

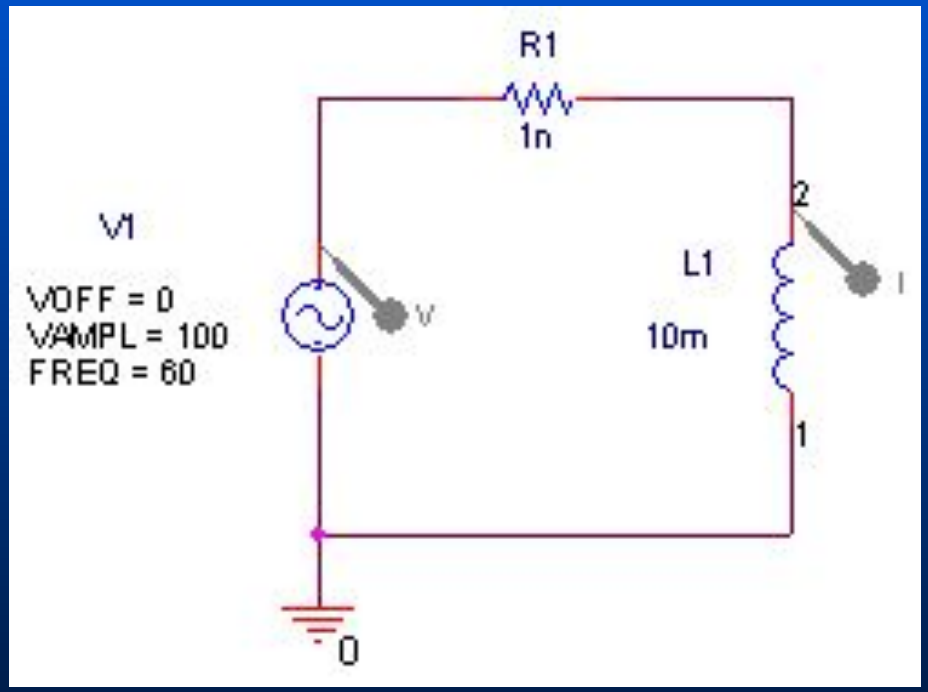
- 전압

$$v = L\frac{di}{dt} = L\frac{d}{dt}I_m \sin\omega t = LI_m \frac{d\sin\omega t}{d\omega t}\frac{d\omega t}{dt}$$

$$= \omega LI_m \cos\omega t = \omega LI_m \sin\left(\omega t + \frac{\pi}{2}\right)$$

- 전압과 전류의 위상: 전류가 90°이 뒤짐(lag)

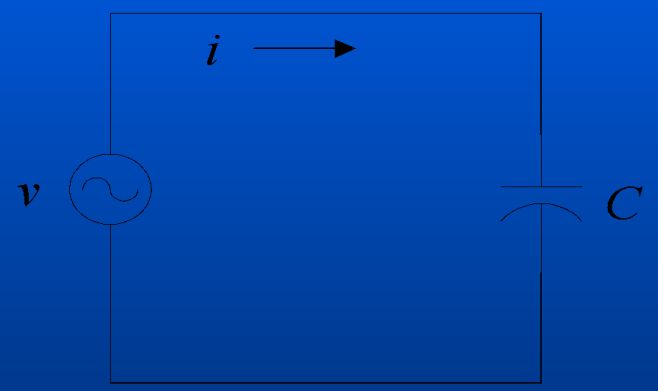
L 회로: OrCAD Simulation

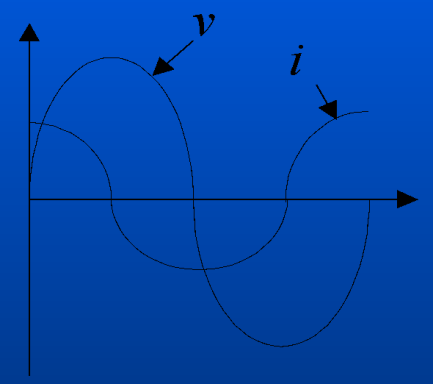


- *Parameter*
  - ✔ *R=1 nΩ*
  - ✔ *L=10mH*
  - ✔ *V1=100 V*
  - ✔ *f=60 Hz*
  - ✔ *Transient Step:*

    0 0.1 ms 50 ms

예비학습: 커패시터(C) 회로





$$\frac{d}{dx}\sin x = \cos x$$

$$\frac{d}{dx}\cos x = -\sin x$$

- 전류

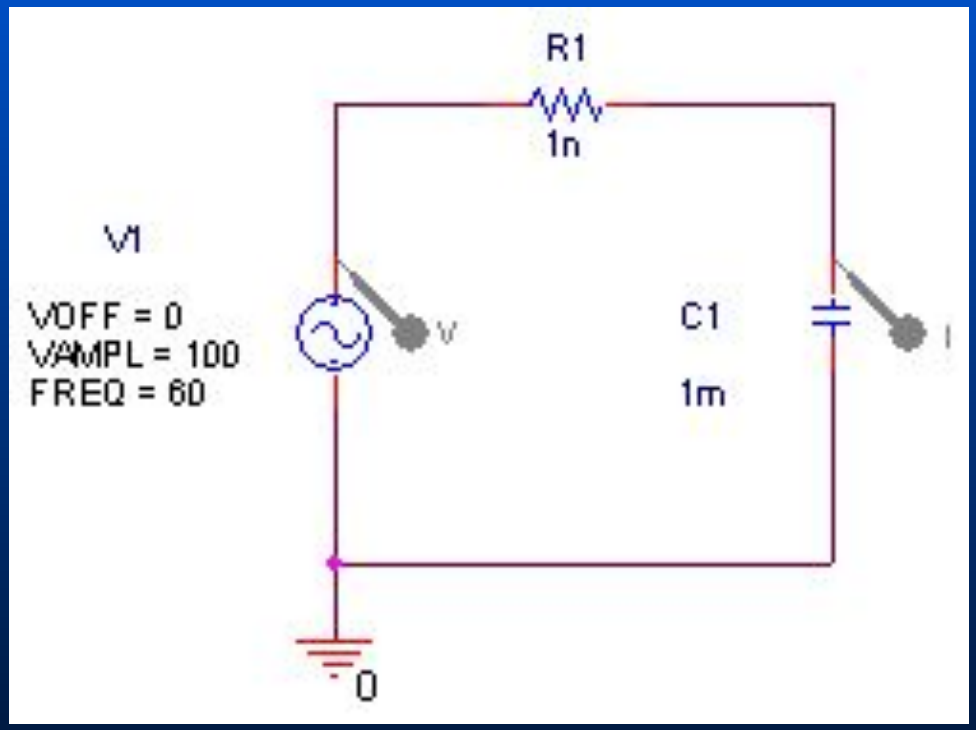
$$Q = Cv$$

$$i = \frac{dQ}{dt} = C\frac{dv}{dt} = C\frac{d}{dt}V_m \sin\omega t = CV_m \frac{d\sin\omega t}{d\omega t}\frac{d\omega t}{dt}$$

$$= \omega CV_m \cos\omega t = \omega CV_m \sin\left(\omega t + \frac{\pi}{2}\right)$$

- 전압과 전류의 위상: 전류가 90° 앞섬(lead)

C 회로: OrCAD Simulation



- *Parameter*
  - ✔ *R=2 Ω*
  - ✔ *C=1mF*
  - ✔ *V1=100 V*
  - ✔ *f=60 Hz*
  - ✔ *Transient Step:*
    0 0.1 ms 50 ms

2.3 인덕터와 커패시터: Inductor

- 인턱터의 저장에너지 (Storage En

$$w(t) = \int i(t)v(t)\,dt = \int i(t)L\frac{di(t)}{dt}dt$$

$$= L\int i(t)di(t) = \frac{1}{2}Li^2(t)\ [J]$$

$$\int 1\,dt = t$$

$$\int x\,dx = \frac{1}{2}x^2$$

$$\int y^{100}\,dy = \frac{1}{101}y^{101}$$

- 평균소비전력: 전류가 주기함수 일 때
  - ✔ 주기전류: 시작=끝

$$P_L = 0[W],\ \ i(t_0+T) = \frac{1}{L}\int_{t_0}^{t_0+T} v_L(t)\,dt + i_0$$

$$i(t_0+T) - i_0 = \frac{1}{L}\int_{t_0}^{t_0+T} v_L(t)\,dt = 0$$

$$v_L(t) = L\frac{di}{dt}$$

$$i = \frac{1}{L}\int_{t_0}^{t_0+T} v_L(t)\,dt + i_0$$

2.3 인덕터와 커패시터: Capacitor

- 캐패시터의 저장에너지 (Storage E

$$\int 1\,dt = t$$

$$\int x\,dx = \frac{1}{2}x^2$$

$$\int y^{100}\,dy = \frac{1}{101}y^{101}$$

$$w(t) = \int v(t)i(t)\,dt = \int v(t)C\frac{dv(t)}{dt}dt$$

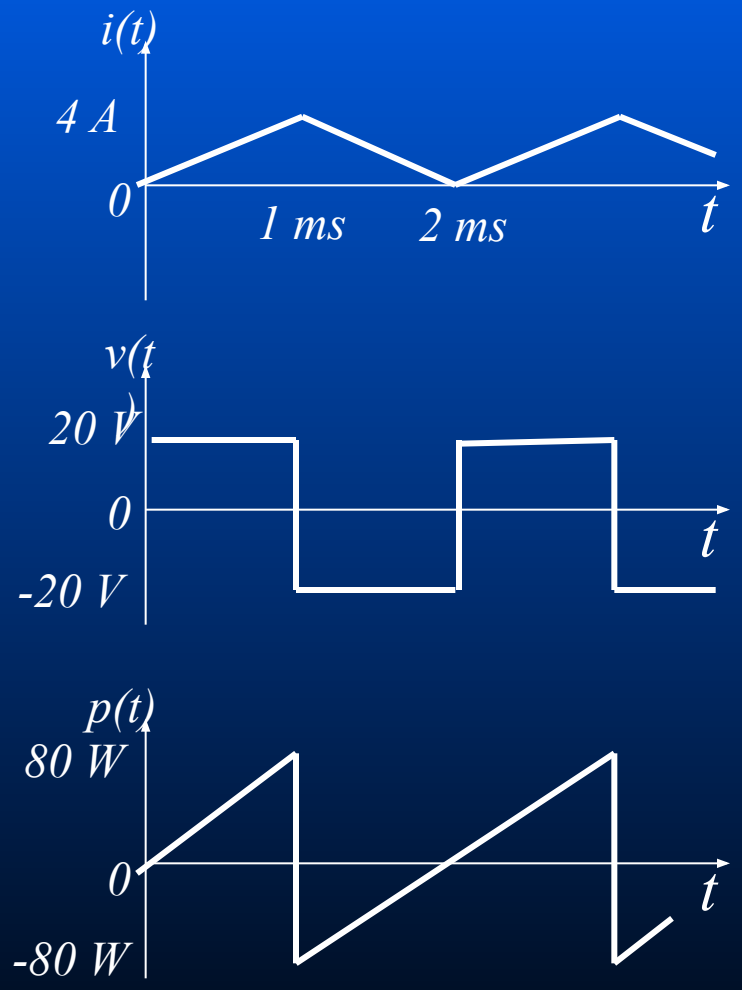
$$= C\int v(t)dv(t) = \frac{1}{2}Cv^2(t)\ [J]$$

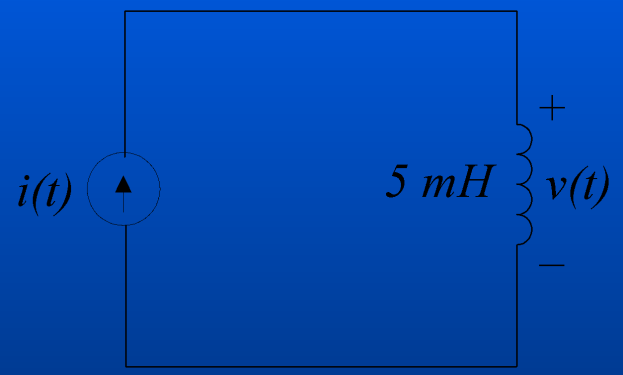
- 평균소비전력: 전압이 주기함수 일때
  - ✔ 주기전압: 시작=끝

$$P_C = 0[W],\ v(t_0+T) = \frac{1}{C}\int_{t_0}^{t_0+T} i_C(t)\,dt + v_0$$

$$v(t_0+T) - v_0 = \frac{1}{C}\int_{t_0}^{t_0+T} i_C(t)\,dt = 0$$

예제 2-2 인덕터의 전력과 전압





- 전압

$$v(t) = L\frac{di}{dt} = 0.005\frac{4-0}{0.001-0} = 20[V]$$

- 전력

$$p(t) = i(t)v(t) = 4 \times 20 = 80[W]$$

$$at\ 1mS$$